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Research Reports

Management Forecasts and Information Asymmetry: An Examination of Bid–Ask Spreads

MARIBETH COLLER* AND TERI LOMBARDI YOHN†

1. Introduction

This paper investigates whether the decision to issue a management earnings forecast is related to information asymmetry in the market for the firm's stock and whether the forecasts reduce the information asymmetry. Theoretical models of the bid–ask spread hold that a portion of the bid–ask arises because of asymmetric information and that specialists widen spreads when they perceive greater information asymmetry.¹ Managers may release a management forecast to provide information that is otherwise held by only a subset of investors and thereby reduce

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¹Copeland and Galai [1983] and Glosten and Milgrom [1985] assume that the market consists of informed and uninformed (liquidity) traders. Specialists fall into the latter category. Kim and Verrecchia [1994] explain that specialists are not thought to do any fundamental analysis, since their duty is to maximize the turnover of capital rather than the value of inventory. Because specialists sustain losses from trading with informed traders, an increase in information asymmetry causes the specialist to widen the bid–ask spread in order to recoup these losses.

the bid–ask spread.² If such forecasts are effective, the bid–ask spread should decrease.

We analyze bid–ask spreads over twelve-month and nine-day windows prior to management earnings forecasts to determine whether forecasting firms have more information asymmetry prior to the forecast than a matched sample of nonforecasting firms. We find that the spreads of forecasting firms are significantly larger over both pre-forecast windows. Examining spreads for forecasters and nonforecasters during the nine days *after* the release of the management forecast, we find no significant differences in spreads.

We also analyze bid–ask spreads for forecasting firms around the management forecast date. We find that forecasting firms have increasing spreads over the twelve months prior to the forecast. There is a temporary increase in spreads the day of and the day after the management forecast; however, spreads in the nine days after the forecast are significantly smaller than spreads in the nine days prior to the forecast.

This evidence on bid–ask spreads helps explain why managers issue forecasts.³ This explanation is consistent with prior empirical findings. For example, Frankel, McNichols, and Wilson [1995] find that firms accessing the capital markets are more likely to disclose earnings forecasts. Skinner [1994], Kasznik and Lev [1995], and Francis, Philbrick, and Schipper [1994] suggest that managers issue forecasts to warn investors about earnings declines. In these cases, managers gain (or avoid losses) from reducing information asymmetry; our results point to reduced spreads as another source of benefit.

The results may also provide insight into how a specialist responds to information releases. Kim and Verrecchia [1994] suggest that a specialist will increase the bid–ask spread around public disclosures to protect himself from traders who are better able to process the public information. Lee, Mucklow, and Ready [1993] find an increase in spreads the day prior to, the day of, and the day after earnings announcements. Our finding of an increase in spreads the day of and day after the management forecast release suggests that, as with earnings announcements, specialists temporarily increase the spread as protection from investors with superior processing abilities. However, consistent with management forecasts being unanticipated, we do not find an increase in spreads immediately prior to the management forecast.⁴

² Managers may also have a direct interest in the bid–ask spread; a firm may attract investors by reducing transactions costs resulting from the specialist's spread.

³ Analytical models provide alternative explanations. Trueman [1986] develops a model which suggests that managers forecast to signal their ability to anticipate future changes in the firm's economic environment. Verrecchia [1983], Darrrough and Stoughton [1990], and Wagenhofer [1990] develop models which suggest that disclosures occur because nondisclosers are assumed by investors to have the worst possible information.

⁴ Kross and Schroeder [1984] show that over 80% of earnings announcements are within three days of the date predicted, while Copeland and Marioni [1972] find that there is no consistent time of year when managers issue earnings forecasts.

2. Research Method

Our analysis involves regressing bid–ask spreads on forecast indicator variables and on variables that proxy for other factors that have been found to affect bid–ask spreads. In cross-sectional analyses of forecasting and nonforecasting firms prior to and after management forecasts, indicator variables are used to capture differences in spreads between the forecasters and the nonforecasters. In time-series analyses of forecasting firms, we examine spreads in the periods before, during, and after the release of a management forecast. In these analyses, indicator variables are used to capture differences in spreads between various time periods.

2.1 SAMPLE DESCRIPTION

We used the Dow Jones News Retrieval Service (*DJNRS*) from 1988 through 1992 to obtain a sample of 520 quarterly earnings forecasts, including point estimates, range estimates, and upper- and lower-bound estimates disclosed by management at least one month before the earnings release date. Forecasts are included only if no other corporate announcements are made in the month prior to the forecast date.⁵ Price and transaction data for 278 observations are available from the Institute for the Study of Security Markets (*ISSM*) tape.⁶

Nonforecasting firms are matched on industry (two-digit *SIC* code), fiscal year-end, exchange listing, and market value of equity. To identify matching firms, *Compustat* firms with a market value of equity within $\pm 5\%$ of each forecasting firm's value are identified. A nonforecasting firm that matches on the other criteria is then found within this group of firms. Matches were found for 179 of the 278 forecasting firms. Most of the sample (58%) consists of fourth-quarter earnings forecasts, with the remainder approximately equally spread over the other three quarters. Two industries (as determined by two-digit *SIC* codes) have more than ten observations: Chemicals/Allied Products (21 forecasts) and Machinery, Nonelectrical (14 forecasts).

2.2 VARIABLE DESCRIPTION

Consistent with previous research, our dependent variable, *SPREAD*, is the average of all bid–ask spreads quoted during each day.⁷

⁵ Daily bid–ask spreads during the month prior to the forecast are an important part of our analyses. We eliminate firms with other announcements during that period because bid–ask spreads may change when new information is released.

⁶ The *ISSM* tape used for this study provides data for firms that trade on the New York Stock Exchange, American Stock Exchange, and regional exchanges. The loss of 242 forecasting firms in gathering the bid–ask spread data is due to the loss of all firms traded through *NASDAQ*.

⁷ To minimize data errors, we apply a filter in which we include only positive spreads less than \$10. Some prior studies (e.g., Stoll [1978]) use the bid–ask spread standardized by the stock price as the dependent variable in regressions. Regressions with the standardized spread as the dependent variable yield results that are qualitatively similar to those reported.

We wish to focus on the portion of the bid–ask spread that is due to asymmetric information, so we need to control for the specialist’s order costs and inventory carrying costs. The order cost component represents the cost of standing ready to trade immediately; the inventory cost portion reflects the cost of maintaining sufficient stock to trade on demand.

Following Demsetz [1968], Tinic [1972], and Stoll [1978], who find a negative relation between trading activity and the bid–ask spread, we include as a control variable the number of transactions (*TRANS*) in a firm’s stock in each day (month).⁸ Following Barnea and Logue [1975], Hamilton [1978], and Stoll [1978], who find that the variability of returns affects the inventory carrying cost component of the spread and thus increases the bid–ask spread, we also include as a control variable the daily (or monthly) variance (*VAR*) of a stock’s returns. Finally, Demsetz [1968], Tinic [1972], Tinic and West [1972], and Benston and Hagerman [1974] find that price is positively related to spreads. Based on these results, we include the average of all individual buy and sell transaction prices (*PRICE*) during each day (month).

In addition to these control variables, we also include a measure for quoted depth. Lee, Mucklow, and Ready [1993] point out that a complete quote includes the best price available for both purchases (the ask) and sales (the bid), as well as the number of shares available at each price (the depth). Specialists can manage asymmetric information risk by using both spreads and depth, so a combination of higher (lower) spread and lower (higher) depth is needed to infer a decrease (increase) in liquidity. Based on this finding, we include in the regressions quoted depth (*DEP*), defined as the average of number of shares in all quotes (the number available at the ask plus the number available at the bid) during each day (month).⁹

In the twelve-month analysis, the dependent and control variables are monthly measures (averages for *SPREAD*, *PRICE*, and *DEP*; totals for *VAR* and *TRANS*) for each of the twelve months prior to the management forecast (months $[-12, -1]$, where month 0 is the month of the forecast). Month -1 is the thirty-day period ending eleven days prior to the forecast date, month -2 is the thirty-day period prior to month -1 , and so on. In the nine-day analysis, the variables are daily measures (averages for *SPREAD*, *PRICE*, and *DEP*; totals for *TRANS* and *VAR*) for each of the nine days prior to or after the forecast (days $[-10, -2]$ or $[+2, +10]$, where day 0 is the day of the forecast) or prior to the earnings announcement (days $[-10, -2]$, where day 0 is the day of the announcement). Following prior research, we use the natural logarithm of all variables (except indicator variables) in the regressions.

⁸ Regressions with the number of shares traded daily as the proxy for trading activity yield results that are qualitatively similar to those reported.

⁹ Regressions run with depth as the dependent variable instead of bid–ask spreads yield results that are qualitatively similar to those reported.

Table 1 provides sample means for spreads and for the control variables, for forecasting and nonforecasting firms for each of three time periods: the nine days prior to, the day of and day after, and the nine days after the management forecast date. Panel B includes means for the nine days prior to the date that the earnings being forecasted are announced. These data support the need to control for factors previously demonstrated to affect the bid-ask spread. *SPREAD* is larger, significant at the .05 level or better using a two-tailed *t*-test, for the forecasters than for the nonforecasters for each of the periods shown in panel A, possibly indicating higher information asymmetry for forecasters both before and after the management forecast. However, previous research has found that *SPREAD* is positively related to *PRICE* and *VAR*, and negatively related to *TRANS* and *DEP*. The significant differences in these variables between the forecasters and the nonforecasters during the various time periods may indicate differences in *SPREAD* based on factors other than information asymmetry.

The indicator variable used in cross-sectional analysis, *FCST*, takes on the value one for forecasting firms, zero otherwise. In time-series analyses, three indicator variables are used. *MONTH* is set equal to zero for month -12 relative to the forecast month, one for month -11, and so on up to eleven for the month immediately prior to the forecast. *MFBEF* takes on the value one for the nine-day period prior to the forecast, zero for the nine-day period after the forecast. In three separate regressions, *DAY* is set equal to one for the day before, the day of, and the day after the forecast, zero for other days in the 21-day period surrounding the forecast.

3. Regression Analysis

3.1 CROSS-SECTIONAL ANALYSIS: FORECASTING VERSUS NONFORECASTING FIRMS

We test for differences in spreads between forecasting and nonforecasting firms by regressing *SPREAD* on *PRICE*, *TRANS*, *VAR*, *DEP*, and *FCST*. The sample includes 179 forecasting and 179 matched nonforecasting firms. The model is:

$$\ln(SPREAD_{i,t}) = \alpha_0 + \alpha_1 \ln(PRICE_{i,t}) + \alpha_2 \ln(TRANS_{i,t}) + \alpha_3 \ln(VAR_{i,t}) + \alpha_4 \ln(DEP_{i,t}) + \alpha_5 FCST_i. \quad (1)$$

Table 2 reports the results of our cross-sectional analysis. During both time periods prior to the forecast, *SPREAD* is positively related to *PRICE* and negatively related to *TRANS* and *DEP*, all at the .01 level, consistent with prior research.¹⁰ The coefficient on *VAR* is not significant at the .10

¹⁰ The relations are consistent with Demsetz [1968], Tinic [1972], Stoll [1978], Barnea and Logue [1975], Hamilton [1978], Tinic and West [1972], and Benston and Hagerman [1974].

TABLE 1
Sample Means for 179 Management Forecasting Firms and
179 Matched Nonforecasting Firms (1988–92)^a

Panel A: Relative to Management Forecast Date			
Time Period	Variable ^b	Forecasters	Nonforecasters
Nine Days Prior to Management Forecast Date (Ending on day -1)	SPREAD***	0.291	0.265
	PRICE***	31.774	28.259
	TRANS***	149.360	179.754
	VAR*	10.941	5.067
	DEP***	36,998	16,397
Day Of and Day After Management Forecast Date (Days 0, +1)	SPREAD**	0.282	0.256
	PRICE**	31.176	28.086
	TRANS***	252.455	176.753
	VAR***	21.305	6.237
	DEP***	27,143	19,511
Nine Days After Management Forecast Date (Beginning on day +2)	SPREAD***	0.281	0.263
	PRICE***	31.024	28.226
	TRANS	176.674	176.215
	VAR	8.190	5.456
	DEP***	29,259	10,625
Panel B: Relative to Earnings Announcement			
Time Period	Variable ^b	Forecasters	Nonforecasters
Nine Days Prior to Earnings Announcement (Ending on day -1)	SPREAD***	0.289	.272
	PRICE***	33.217	30.075
	TRANS***	167.602	141.328
	VAR	8.499	7.213
	DEP	17,754	23,542

^aMatched on industry, fiscal year-end, exchange listing, and market value of equity.
^bSPREAD is the average of all bid-ask spreads quoted during each day. PRICE is the average of all transaction prices during each day. TRANS is the number of transactions per day. VAR is the daily variance of returns. DEP is the average number of shares in all quotes during the day.
*, **, ***Significantly different for forecasters vs. nonforecasters at .10, .05, and .01 levels, respectively (two-tailed *t*-test).

level. In both pre-forecast periods, the coefficient on *FCST* is positive and significant at the .01 level. Forecasting firms have a significantly higher level of information asymmetry than nonforecasting firms during the twelve months and the nine days prior to the forecast release.

If the release of a forecast reduces information asymmetry in the market for the firm's stock, the difference in spreads between forecasters and nonforecasters in the pre-forecast period should be reduced in the period following the forecast. To test this prediction, we re-estimate equation (1) using days [+2,+10], where day 0 is the forecast date. Results in table 2 indicate that the coefficient on *FCST* is not significantly different from zero (*t* = 0.14). The significantly higher spreads found prior to management forecasts for forecasting firms are not found in the nine days after the forecast.

TABLE 2

Forecasting Firms and Matched Nonforecasting Firms: Coefficient Estimates (t-Statistics), 358 Firms (1988–92)

$$\text{Model: } \ln(\text{SPREAD}_{i,t}) = \alpha_0 + \alpha_1 \ln(\text{PRICE}_{i,t}) + \alpha_2 \ln(\text{TRANS}_{i,t}) + \alpha_3 \ln(\text{VAR}_{i,t}) + \alpha_4 \ln(\text{DEP}_{i,t}) + \alpha_5 \text{FCST}_i$$

Panel A: Monthly Spreads for Twelve Months Prior to Management Forecast^a

Time Period	α_0	α_1	α_2	α_3	α_4	α_5	Adj. R^2
$-12 \leq t \leq -1$	-1.613	0.294	-0.064	-0.001	-0.040	0.030	0.351
$t = 0$ is month of forecast	(-49.7)***	(37.6)***	(-11.6)***	(-0.59)	(-5.06)***	(2.34)***	

Panel B: Daily Spreads for Nine Days Prior to Management Forecast^b

$-10 \leq t \leq -2$	-1.986	0.336	-0.082	0.002	-0.009	0.032	0.426
$t = 0$ is day of forecast	(-59.7)***	(45.7)***	(-16.4)***	(0.16)	(-3.69)***	(2.91)***	

Panel C: Daily Spreads for Nine Days After Management Forecast^b

$+2 \leq t \leq +10$	-1.927	0.329	-0.094	-0.0004	-0.009	0.001	0.409
$t = 0$ is day of forecast	(-36.4)***	(31.1)***	(-13.2)***	(-0.21)	(-2.54)***	(0.14)	

Panel D: Daily Spreads for Nine Days Prior to Earnings Announcement^b

$-10 \leq t \leq -2$	-1.945	0.328	-0.077	-0.001	-0.011	-0.001	0.454
$t = 0$ is day of earnings announcement	(-57.0)***	(49.2)***	(-16.5)***	(-0.30)	(-3.99)***	(-0.13)	

^a i = firm; t = month; $\text{FCST}_i = 1$ if firm i is a forecasting firm, 0 otherwise. *SPREAD* is the average of all bid-ask spreads quoted during each month. *PRICE* is the average of all transaction prices during each month. *TRANS* is the number of transactions per month. *VAR* is the monthly variance of returns. *DEP* is the average number of shares in all quotes during the month.

^b i = firm; t = day; $\text{FCST}_i = 1$ if firm i is a forecasting firm, 0 otherwise. *SPREAD* is the average of all bid-ask spreads quoted during each day. *PRICE* is the average of all transaction prices during each day. *TRANS* is the number of transactions per day. *VAR* is the daily variance of returns. *DEP* is the average number of shares in all quotes during the day.

*, **, ***Significant at .10, .05, and .01 levels, respectively (one-tailed).

To test whether the decreased spread differential is a temporary effect, we examine the difference in spreads during the nine-day period prior to the earnings announcement date (at least thirty days after the management forecast). Again, the coefficient on *FCST* is not significant ($t = -0.13$). It thus appears that the forecast reduces differences in information asymmetry, at least until the earnings announcement. These results support the Conference Board assertions documented in Ajinkya and Gift [1984] that managers forecast earnings to reduce unequal access to private information.

3.2 TIME-SERIES ANALYSIS: FORECASTING FIRMS

By examining bid-ask spreads of the 278 forecasting firms around the date of the management forecast, we can test whether there is a change in information asymmetry during the twelve months prior to the forecast that might prompt the forecast release. We use monthly data with MONTH_i to specify the month prior to the forecast date:

$$\ln(SPREAD_{i,t}) = \alpha_0 + \alpha_1 \ln(PRICE_{i,t}) + \alpha_2 \ln(TRANS_{i,t}) + \alpha_3 \ln(VAR_{i,t}) + \alpha_4 \ln(DEP_{i,t}) + \alpha_5 MONTH_t. \quad (2)$$

A positive coefficient on $MONTH_t$ would indicate that the forecasting firms experienced increasing information asymmetry over the twelve months prior to the release of the forecast.

The coefficient estimates are reported in table 3. The coefficient on $MONTH_t$ is positive and significant (at the .05 level, $t = 1.93$), suggesting that a perception of an increase in information asymmetry may be a possible motivation for issuing a management forecast.¹¹

To test whether the release of a management forecast reduces information asymmetry in the market for the forecasting firm's stock, we estimate model (3):

$$\ln(SPREAD_{i,t}) = \alpha_0 + \alpha_1 \ln(PRICE_{i,t}) + \alpha_2 \ln(TRANS_{i,t}) + \alpha_3 \ln(VAR_{i,t}) + \alpha_4 \ln(DEP_{i,t}) + \alpha_5 MFBEF_t. \quad (3)$$

Daily measures for each variable are included in the regression for the periods $[-10, -2]$ and $[+2, +10]$, where day 0 is the day of the management forecast.

Table 3 also presents these regression results. The coefficient on $MFBEF_t$ is positive and significant at the .05 level, indicating that the forecasting firms experience a decrease in spreads from the nine days before the forecast to the nine days after. This suggests that management forecasts reduce information asymmetry in the market for the firm's stock.

We also examine the effect of the management forecast on bid-ask spreads during the three days surrounding the forecast date. The model is:

$$\ln(SPREAD_{i,t}) = \alpha_0 + \alpha_1 \ln(PRICE_{i,t}) + \alpha_2 \ln(TRANS_{i,t}) + \alpha_3 \ln(VAR_{i,t}) + \alpha_4 \ln(DEP_{i,t}) + \alpha_5 DAY_t. \quad (4)$$

The observations include the twenty-one days surrounding the management forecast (days $[-10, +10]$). We run three separate regressions to determine whether spreads widen on any of the three days surrounding the management forecast. In the first regression, $DAY = 1$ for the day prior to the forecast; $DAY = 1$ for the forecast date in the second; and $DAY = 1$ for the day after the forecast in the third regression. In all regressions, $DAY = 0$ for the eighteen days of the nonevent period.¹²

¹¹ We also ran regressions to determine whether there is a particular month prior to the forecast in which a sudden increase in bid-ask spread occurs. We were unable to identify any particular point when such a shift occurs.

¹² Based on research examining bid-ask spreads around earnings announcements, the $[-1, 0, +1]$ period was determined a priori to be the most likely period in which the spread may widen. To isolate any changes on each of these days from the other eighteen days, the two event days not being tested in a given regression are excluded. An alternative specification was run in which these two days were included in the nonevent period. The

TABLE 3
278 Forecasting Firms: Coefficient Estimates (t-Statistics), 1988–92

Panel A: Twelve Months Prior to Management Forecast^a

$$\ln(\text{SPREAD}_{i,t}) = \alpha_0 + \alpha_1 \ln(\text{PRICE}_{i,t}) + \alpha_2 \ln(\text{TRANS}_{i,t}) + \alpha_3 \ln(\text{VAR}_{i,t}) + \alpha_4 \ln(\text{DEP}_{i,t}) + \alpha_5 \text{MONTH}_t$$

Time Period	α_0	α_1	α_2	α_3	α_4	α_5	Adj. R^2
$-12 \leq t \leq -1$	-1.706	0.298	-0.053	-0.002	-0.043	0.004	0.333
$t = 0$ is month of forecast	(-39.6)***	(30.4)***	(-8.36)***	(-1.16)	(-4.52)***	(1.93)**	

Panel B: Nine Days Before vs. Nine Days After Management Forecast^b

$$\ln(\text{SPREAD}_{i,t}) = \alpha_0 + \alpha_1 \ln(\text{PRICE}_{i,t}) + \alpha_2 \ln(\text{TRANS}_{i,t}) + \alpha_3 \ln(\text{VAR}_{i,t}) + \alpha_4 \ln(\text{DEP}_{i,t}) + \alpha_5 \text{MFBEF}_t$$

$-10 \leq t \leq -2$	-2.047	0.353	-0.083	0.003	-0.008	0.020	0.475
and $+2 \leq t \leq +10$	(-41.8)***	(35.2)***	(-11.7)***	(1.60)*	(-2.48)***	(1.76)**	
$t = 0$ is day of forecast							

^a i = firm; t = month; $\text{MONTH} = 0$ for month -12 relative to the forecast month, 1 for month -11, 2 for month -10, and so on up to a value of 11 for the month immediately prior to the forecast month. SPREAD is the average of all bid-ask spreads quoted during each month. PRICE is the average of all transaction prices during each month. TRANS is the number of transactions per month. VAR is the monthly variance of returns. DEP is the average number of shares in all quotes during the month.

^b i = firm; t = day; $\text{MFBEF} = 1$ if $-10 \leq t \leq -2$ (the period before the forecast), 0 otherwise. SPREAD is the average of all bid-ask spreads quoted during each day. PRICE is the average of all transaction prices during each day. TRANS is the number of transactions per day. VAR is the daily variance of returns. DEP is the average number of shares in all quotes during the day.

*, **, ***Significant at .10, .05, and .01 levels, respectively (one-tailed).

Results of the daily regressions are presented in table 4. The coefficient on DAY is positive and significant at the .01 level on day zero and at the .10 level on day +1. The results suggest that spreads are significantly larger on the day of and the day after management forecasts than on the other nineteen days examined. The results are consistent with the theoretical model in Kim and Verrecchia [1994] and with increases in spreads on the days surrounding earnings announcements found in Lee, Mucklow, and Ready [1993].

4. Conclusions

This paper examines bid-ask spreads around management forecasts of quarterly earnings to gain insight into why managers issue forecasts and how these forecasts affect information asymmetry. The findings support the view that managers release forecasts to reduce information

results for day 0 do not change from those reported. However, when days $[-1,0]$ are included in the nonevent period, the significance of the dummy variable for day +1 drops to 0.13 (one-tailed). Other regressions, not reported here, show no significant coefficients on DAY for any other day of the 21-day period.

TABLE 4

278 Forecasting Firms for Three Days Around Management Forecasts: Coefficient Estimates (t-Statistics), 1988–92

$$\ln(\text{SPREAD}_{i,t}) = \alpha_0 + \alpha_1 \ln(\text{PRICE}_{i,t}) + \alpha_2 \ln(\text{TRANS}_{i,t}) + \alpha_3 \ln(\text{VAR}_{i,t}) + \alpha_4 \ln(\text{DEP}_{i,t}) + \alpha_5 \text{DAY}_t$$

Model ^a	α_0	α_1	α_2	α_3	α_4	α_5	Adj. R^2
$\text{DAY} = 1$ for $t = -1$ (Days 0, +1 excluded)	-2.264 (-74.8)***	0.412 (61.7)***	-0.074 (-16.7)***	0.003 (2.30)**	-0.006 (-2.61)***	0.132 (0.71)	0.447
$\text{DAY} = 1$ for $t = 0$ (Days -1, +1 excluded)	-2.260 (-75.1)***	0.411 (61.8)***	-0.075 (-17.0)***	0.003 (2.00)**	-0.005 (-2.49)***	0.044 (2.37)***	0.448
$\text{DAY} = 1$ for $t = +1$ (Days -1, 0 excluded)	-2.259 (-74.8)***	0.411 (61.6)***	-0.074 (-16.7)***	0.004 (2.34)***	-0.005 (-2.55)***	0.024 (1.28)*	0.446

^a i = firm; t = day; $-10 \leq t \leq +10$; $t = 0$ for day of management forecast. $\text{DAY} = 1$ for test day, 0 for all other days. SPREAD is the average of all bid–ask spreads quoted during each day. PRICE is the average of all transaction prices during each day. TRANS is the number of transactions per day. VAR is the daily variance of returns. DEP is the average number of shares in all quotes during the day.

*, **, ***Significant at .10, .05, and .01 levels, respectively (one-tailed).

asymmetry. Forecasting firms have greater information asymmetry (as measured by bid–ask spreads) than nonforecasters both during the twelve months and nine days preceding the forecast. This difference in spreads is eliminated immediately after the management forecast. There is no significant difference in the spreads just prior to the subsequent earnings announcement, indicating that the forecast effect persists for some time.

Increased information asymmetry may prompt management to issue a forecast; forecasting firms experience an increase in bid–ask spreads during the twelve months prior to the forecast. Bid–ask spreads increase the day of and the day after the management forecast date; however, the spread for the nine days after the forecast is significantly lower than the spreads prior to the forecast release. This suggests that management forecasts are effective in reducing information asymmetry in the market for the firm’s stock.

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